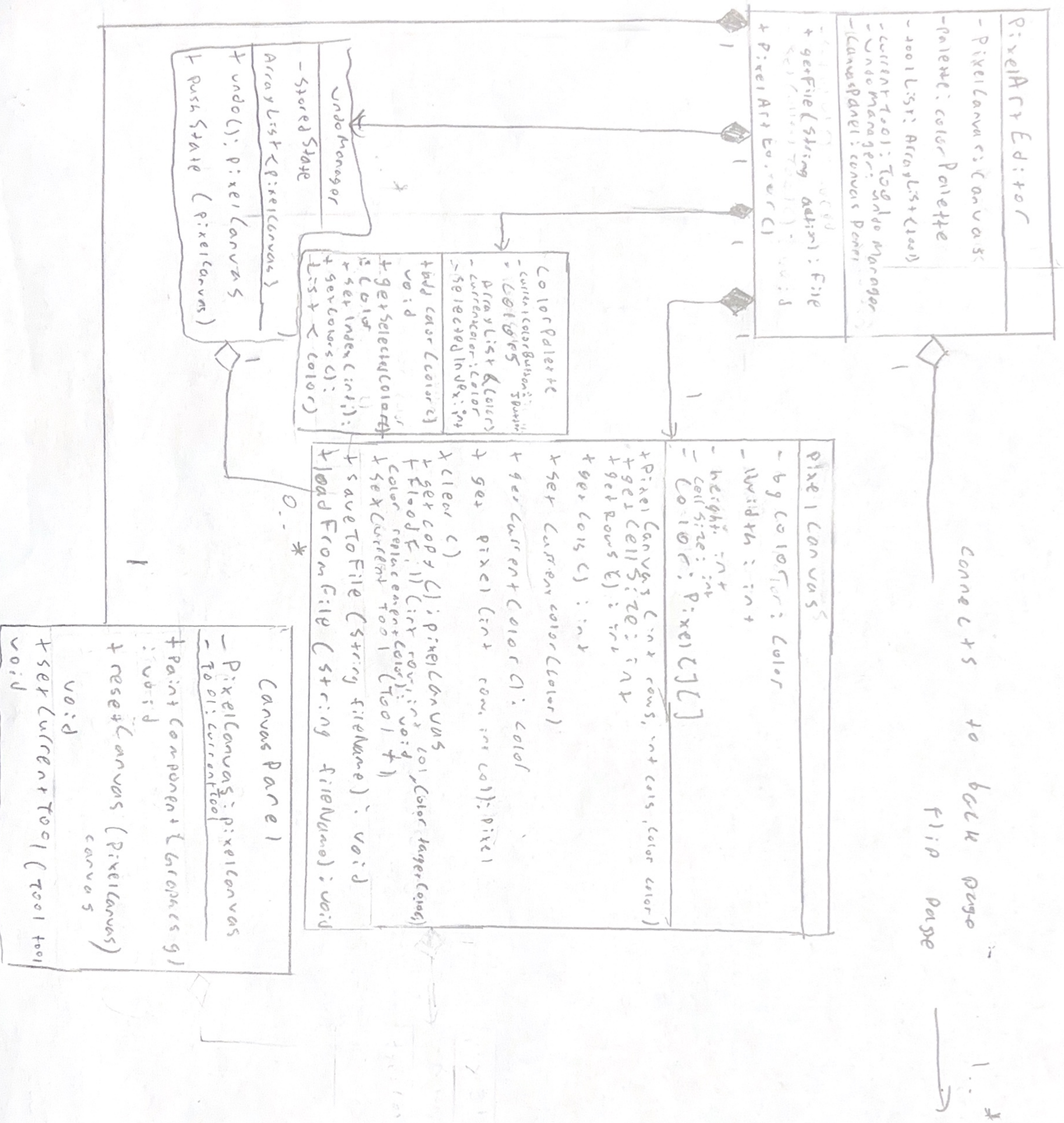


UML Class Diagram



Undo Tool

- Palette: Color palette
+ get Name(): String
+ undo Tool (UndoManager)
+ get Name(): String
+ on Click (PixelCanvas canvas,
int row, int col, Color color)
+ on Release: void

Tool

+ get Name(): String
+ on Click (PixelCanvas canvas, int row, int col): void
+ on Release (PixelCanvas canvas, int row, int col): void
+ on Drag (PixelCanvas canvas, int row, int col): void

Pencil Tool

- Palette: Color palette
- last row: int
- last col: int
+ get Name(): String

+ on Click (PixelCanvas canvas,
int row, int col, Color color): void
+ on Release: void
+ on Drag (PixelCanvas canvas,
int row, int col, Color color)
void

Eraser Tool

- Color by color
- last row: int
- last col: int
+ get Name(): String

+ on Click (PixelCanvas canvas,
int row, int col, Color color)
void
+ on Drag (PixelCanvas canvas,
int row, int col, Color color)
void
+ on Release: void

Fill Tool

- Palette: Color palette
+ get Name(): String
+ on Release: void
+ on Click (PixelCanvas canvas,
int row, int col, Color color)

+ on Drag (PixelCanvas canvas,
int row, int col, Color color,
Color by color): void

Rectangle Tool

- Palette: Color palette
- start row: int
- start col: int
+ get Name(): String
- current row: int
- current col: int

+ on Click (PixelCanvas canvas, int row,
int col, Color color, Color by color)
void
+ on Release (PixelCanvas canvas, int row, int col): void
+ on Drag (PixelCanvas canvas, int row,
int col, Color color, Color by color)
void

Clear Tool

+ get Name(): String
+ on Click: void
+ on Drag: void
+ on Release: void

UI Layout

Pixel Art

File

Clear

Tools

Color Palette

Selected Color

Pencil

Eraser

Undo

Fill

Rectangle

has
save
and
load

PixelCanvas

Brainstorming

Features

- Tool (interface)
 - PencilTool (implements tool) → Lets user colour pixels by clicking or dragging
 - EraseTool (implements tool) → remove pixel color dragging
 - FillTool (implements tool) → recursive flood-fill
 - RectangleTool (implements tool) → startRow, startCol
- PixelArtEditor (main GUI window)
 - ColorPalette (composition to PixelArtEditor) → manages available colors
 - Includes arraylist of colors, colorIndex, currentColor
 - UndoManager (composition to PixelArtEditor)
 - Includes arraylist of states
 - pushState → saves backup of canvas so when it is called can undo
 - FileManager (composition to PixelArtEditor) to previous copy
 - save and load
- PixelCanvas → represents the drawing grid → entire drawing
 - includes a 2D array of pixel board
 - includes all different tools
 - Clear canvas, get and set pixel colors
- Pixel
 - Single pixel in PixelCanvas
 - store, change colour

• UndoTool → calls UndoManager

Custom features: UndoTool, RectangleTool